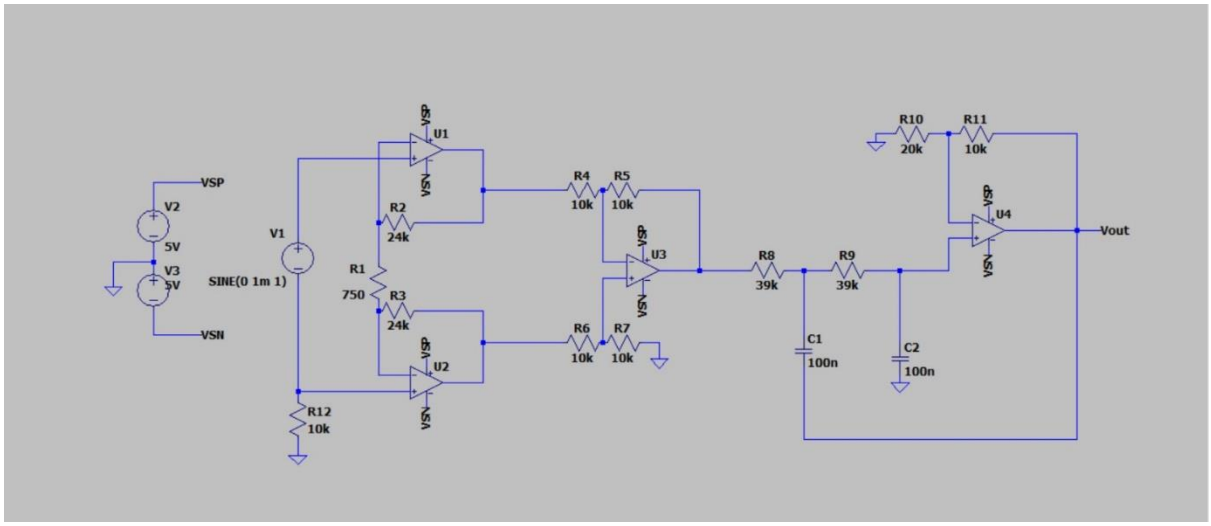
Proiect DEEA

R1 = 750

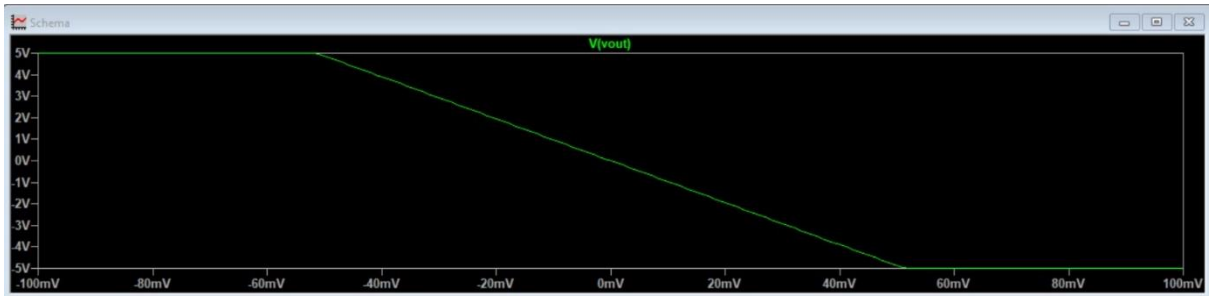
R2 = R2 = 24k

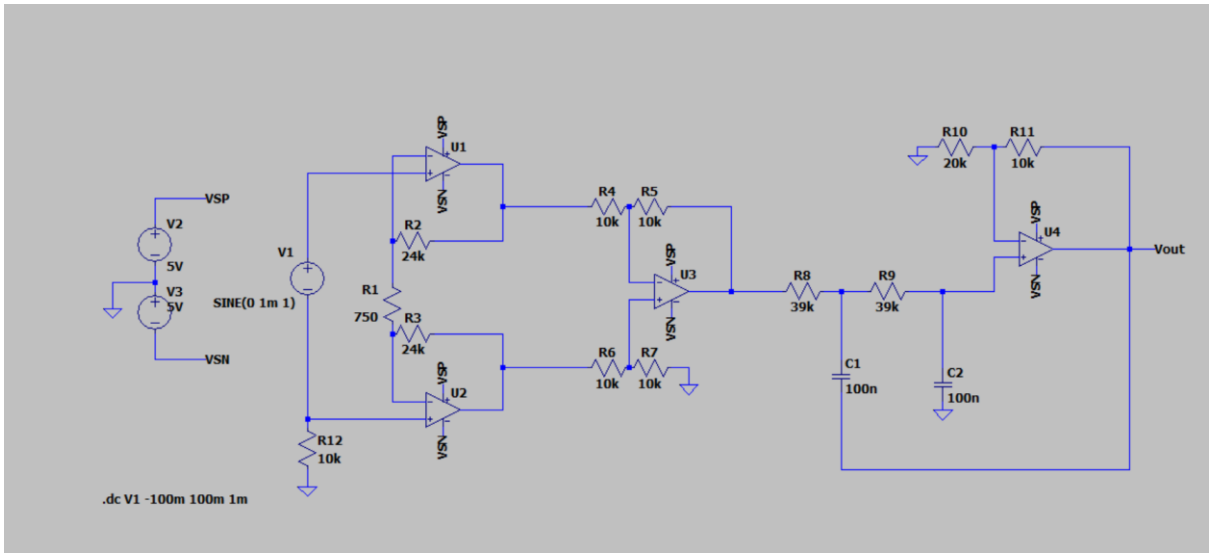
R8 = R9 = 39k

1.



2.





A = (1 + (24k + 24k) / 750) * (-10k / 10k) * (1 + 10k / 20k) = -97.5

Domeniul tensiunii de ieșire este [-5V; 5V]

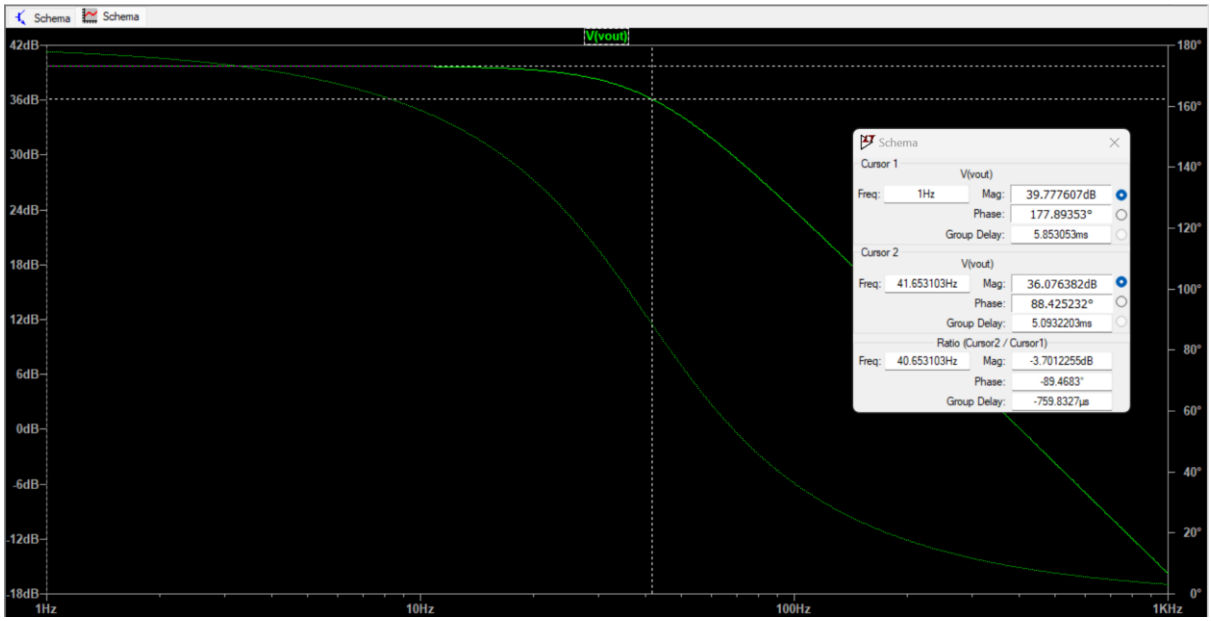
Vin1= -5 / | -97.5 | = -0.05128 V

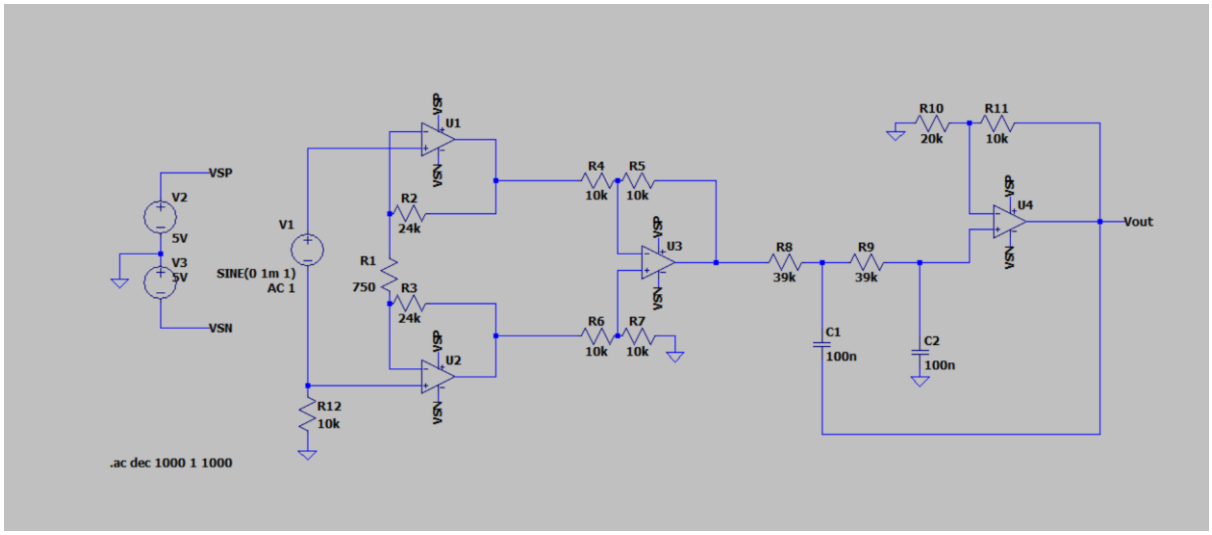
Vin2= 5 / | -97.5 | = 0.05128 V

Domeniul tensiunii este [-51,28mV; 51,28mV]



3.





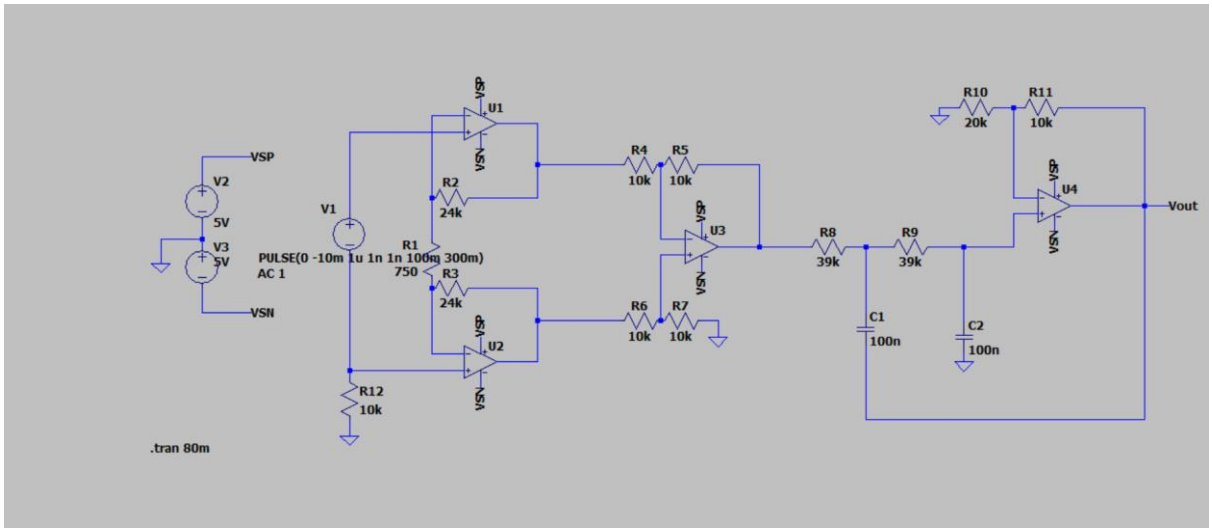
Frecvența de -3.7 dB rezultată din simulare este aproximaiv f-3.7dB = 41,65Hz

4.

Am folosit convenția care măsoară timpul de creștere între 10% (97mV) și 90% (0,87V) din amplitudinea comutației (0,975V).

Timpul de creștere obținut este de aproximativ 8,9 ms.





5.1

Vim = 20

Vom = 4

f-3dB = 2000

Domeniul tensiunii de intrare este [-20; 20] mV

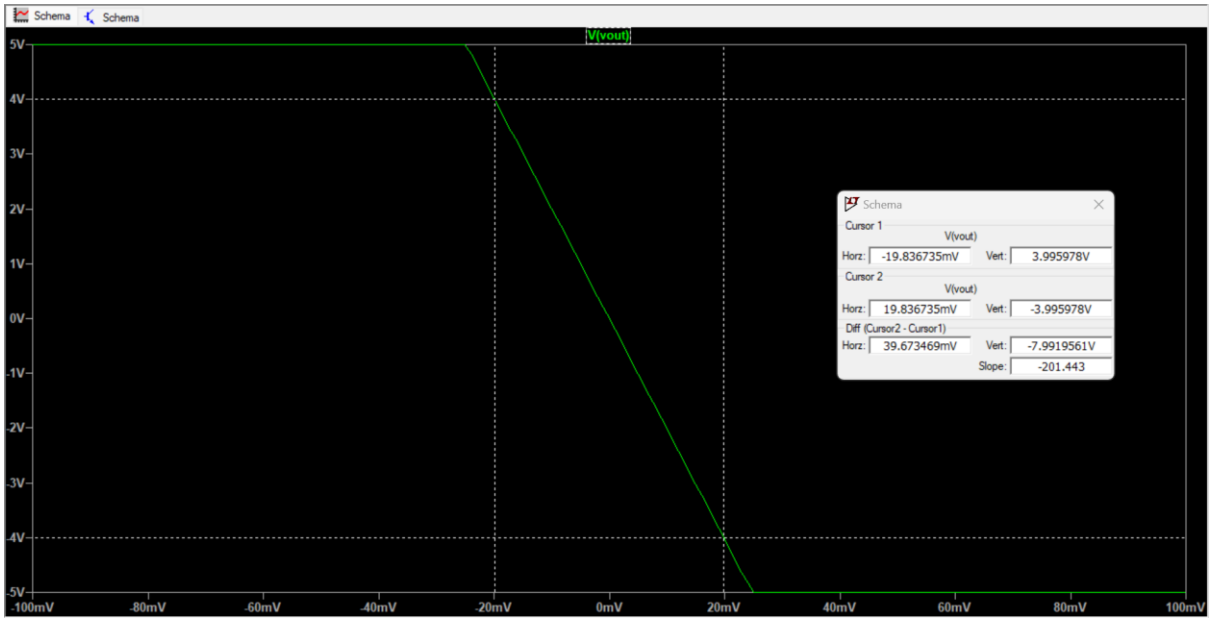
Domeniul tensiunii de ieșire este [-4; 4] V

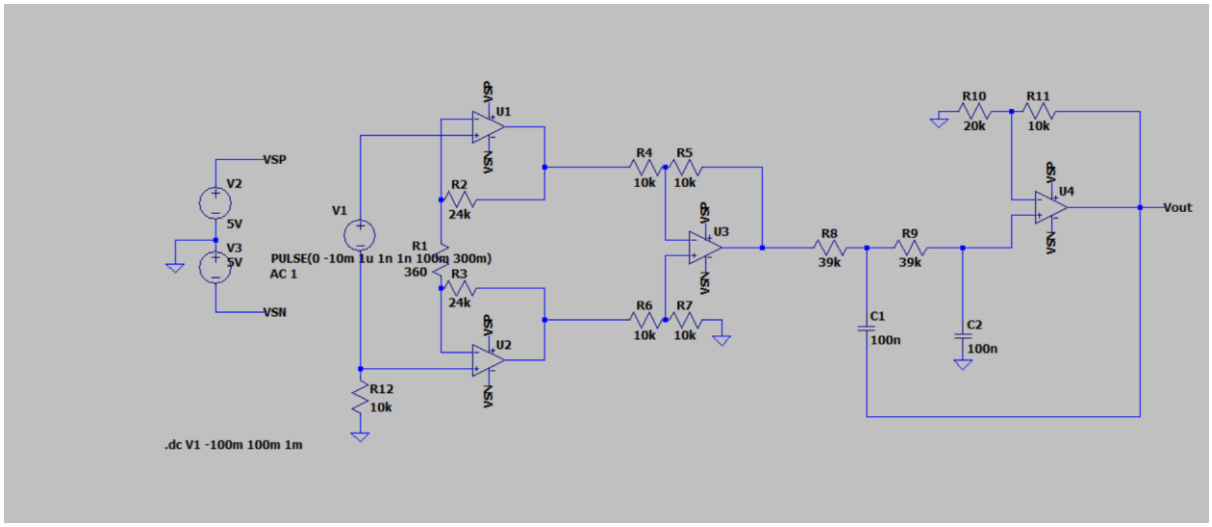
A = ΔVom / ΔVim = (4 – (-4)) / (0,02 – (-0,02)) = 200

-200 = (1 + (24k + 24k) / R1) * (-10k / 10k) * (1 + 10k / 20k)

-200 = -3/2 * (1 + 48k / R1)

R1 = 357Ω => R1 = 360Ω(conform standard E24)





5.2

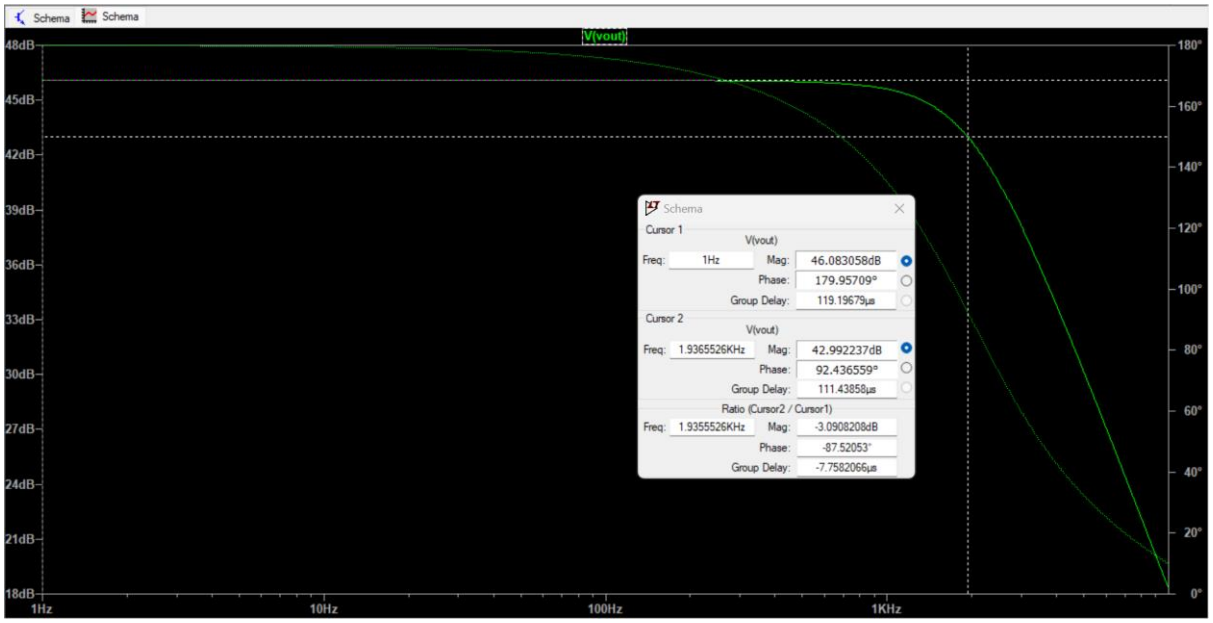
f-3dB = 2000Hz

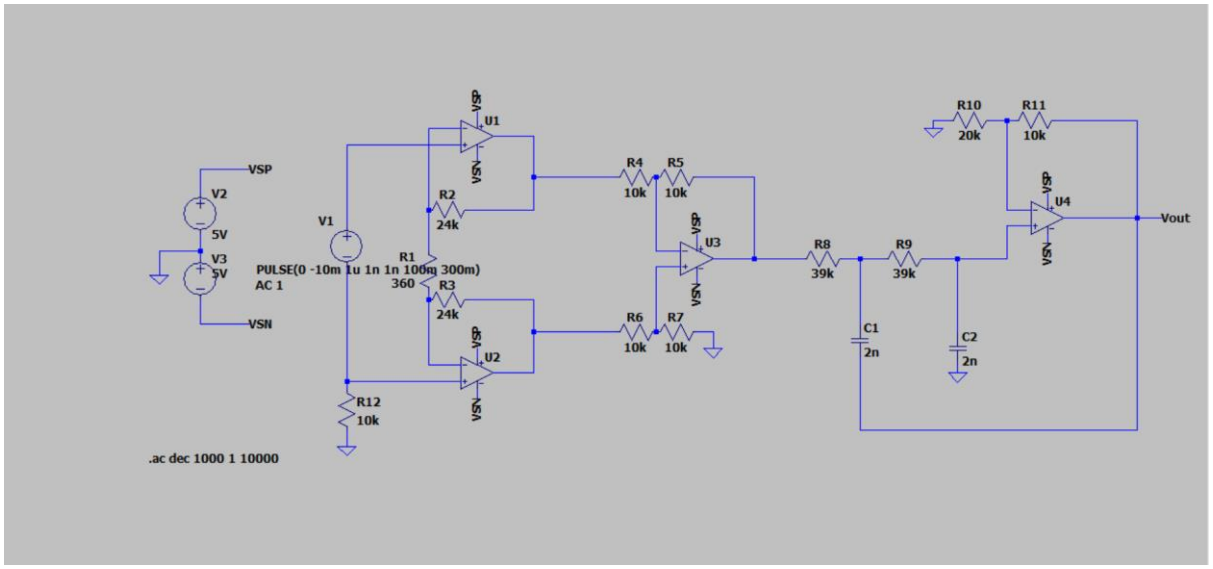
f0 = 41,65Hz

f-3dB / f0 = 48.01

Frecvența de -3dB este proporțională cu 1 / √R8∗R9∗C1∗C2

C1 = C2 = 100 nF / 48.01 = 2.08 nF => C1 = C2 = 2000pF = 2nF(conform standard E24)


Schema
Schema
V(vout)
48dB
45dB
42dB
39dB
36dB
33dB
30dB
27dB
24dB
21dB
18dB
180°
160°
140°
120°
100°
80°
60°
40°
20°
0°
1Hz
10Hz
100Hz
1KHz
Schema
Cursor 1
V(vout)
Freq: 1Hz
Mag: 46.083058dB
Phase: 179.95709°
Group Delay: 119.19679µs
Cursor 2
V(vout)
Freq: 1.9365526KHz
Mag: 42.992237dB
Phase: 92.436559°
Group Delay: 111.43858µs
Ratio (Cursor2 / Cursor1)
Freq: 1.9355526KHz
Mag: -3.0908208dB
Phase: -87.52053°
Group Delay: -7.7582066µs


VSP
V2
5V
V3
5V
VSN
V1
PULSE(0 -10m 1u 1n 1n 100m 300m)
AC 1
R1
360
R2
24k
R3
24k
R4
10k
R5
10k
R6
10k
R7
10k
R8
39k
R9
39k
R10
20k
R11
10k
R12
10k
C1
2n
C2
2n
U1
U2
U3
U4
Vout
.ac dec 1000 1 10000